

Microcontroller Development Boards

Congratulations for your excellent article on 'Microcontroller Development Boards' in October 2013 issue.

I am designing microcontroller-based systems for the last 30 years, initially with 8085 and then with AT89C51 series microcontrollers, and have developed many products using Assembly language.

Now I want to switch over to ARM and want to write software in Assembly as well as in C/C++. My basic requirements are 2k RAM or more, two UART, RTC and in-built line/graphic LCD display controller.

Could you please suggest me some controller series which are easily

available and inexpensive, for which I should be able to get all the support easily including programmer, compiler, debugger and training.

Debasish Das

United Automation, Kolkata

EFY: It is good to hear that you found the article useful. Based on your requirements, I think STM32 series will be most suitable for you. STM32 is a product of STMicroelectronics which has good presence throughout the country and so the controller is easily available. The chips are low cost and their development boards are easily available. You can buy Discovery demo board that will only cost you around ₹ 450 to start with. You can even attend some seminars of ST and get this board free of cost. The board has all I/Os open for development.

Toll chain for the series is also good and easily available. Atollic will be the best compiler but you can also go with IAR and Keil. Programming tools are available on the website of STMicroelectronics. You can use JTAG, SWD or other methods to program the chips from the series.

12V SMPS

I am a regular reader of EFY and it is a very good magazine. Special thanks to EFY for publishing the 'Simple 12V, 1A SMPS' circuit in November 2013 issue.

I want to make some modifications in this circuit, for which I need your help. I want to connect multiple power LEDs using this SMPS circuit, but that will require higher current (5A). Kindly let me know how do I increase current capacity while keeping the same voltage?

I do not have much idea about fly-back transformer, so please explain or give me the links where I can get relevant information. And I am very much confused about this transformer, the use of eight pins in some transformers, why there are four input terminals and how to make use of them in the circuit, etc. Could you tell me where I will get in-depth information about different types of transformers?

Amit Kolekar

Through e-mail

The author Ashwini Vishwakarma replies: I am glad that you liked the article. I would love to help you, but your questions are not directly related to the article at all. They are general SMPS questions and the scopes are very broad.

In order to get the answers to such broad questions, you may refer the following books:

High-Frequency Switching Power Supplies by George Chrysis and Switching Power Supply Design by Abraham Pressman

The link to purchase the books:

http://www.amazon.com/Switching-Power-Supply-Design-3rd/dp/0071482725/ref=sr_1_2?ie=UTF8&qid=1385804808&sr=1-2&keywords=smps

I learnt everything from these two books. Go through these books thoroughly and I am sure you will get the answers from them.

My original article had both these books mentioned at the end of the article in the 'Reference' section but that was deleted by EFY. I would request EFY to keep the reference sections in future articles, as they can be useful for readers. **EFY:** Thanks for the suggestion! The reference section is normally removed from the articles to save space and make the articles more appealing for general reading in the magazine.

'Spot An Error' Award Winners

□ In the circuit idea 'ASK Communication Trainer' published in December 2013 issue, the value of VR1 is mistakenly printed as 4.7 kilo-ohm in the parts list instead of 10 kilo-ohm.

Praveen S. Javali

□ In the circuit idea 'ASK Communication Trainer' published in December 2013 issue, the value R4 is mistakenly printed as 150-ohm in the parts list instead of 100-ohm.

Pradeep S.J.

From electronicsforu.com: Circuits

□ I want to interface 'Environment Monitoring System Using Arduino' project published in August 2013 issue with GSM and send the data signals through GSM.

Rakesh Padashetty

Through e-mail

The author Ronie Adhiraj Ghosh replies: For interfacing with GSM, remove the LCD display and connect it with GSM module through RS-232 port and send the data via AT commands to your mobile SIM number.

□ In the 'Sensitive LPG Leakage Alarm' circuit published in April, 2013 issue, whenever I connect it to 12V supply, the LED glows and IC 4060 gets heated up. The sensor is not detecting gas leakage. Please help.

Saptarshi Mandal

Through e-mail

Which software is used to draw circuit and PCB layout in the above circuit?

Krishna Prasad

Through e-mail

EFY: Please check proper connections in the circuit and also check the test points (TP0 through TP4) as per the voltage listed in the table. We used gEDA software for designing the circuits and PCB layout.